

LESSON 1 QUIZ

The Isolette

Name: _____

Date: _____

How to play: For each question, put an X in the box next to the best answer (one answer per question). Get **6 of 8** right to earn the **hatching chick** badge. Take your time - real doctors think before they answer!

1. What is the very first big problem a premature baby faces after birth?

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- A. Learning to smile
- B. Staying warm
- C. Growing hair
- D. Opening its eyes

2. What does the skin probe taped to the baby's belly do?

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- A. It feeds the baby
- B. It measures the baby's skin temperature so the isolette can adjust its warmth automatically
- C. It plays heartbeat sounds
- D. It measures how much the baby weighs

3. Why can't premature babies warm themselves by shivering?

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- A. Shivering is something you have to learn in school
- B. Their muscles are too small and weak, and newborns don't really shiver anyway
- C. The isolette doesn't allow it
- D. Shivering only works in cold countries

4. What is brown fat for?

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- A. Making heat without shivering
- B. Helping the baby float
- C. Storing extra milk
- D. Making the skin darker

5. A wet baby loses heat fastest by which route?

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- A. Radiation
- B. Conduction
- C. Evaporation
- D. Imagination

6. Why is the air in an isolette kept extra humid for the tiniest babies?

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- A. Because babies like the smell of rain
- B. Their skin is so thin that water escapes through it, taking heat along – moist air slows that down
- C. To keep the plastic from cracking
- D. To help their hair grow

7. What is kangaroo care?

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- A. A special bouncy exercise for babies
- B. Feeding the baby in a pouch-shaped pillow
- C. Holding the baby skin-to-skin on a parent's bare chest
- D. A type of baby blanket from Australia

8. Getting cold is dangerous for a preemie because...

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- A. cold babies cry too loudly
- B. the baby burns up sugar and oxygen trying to make heat, and every other problem gets worse
- C. the isolette might rust
- D. actually, it isn't dangerous – babies don't mind the cold

Score: _____ / 8

Badge earned? YES / NOT YET (need 6)

LESSON 2 QUIZ

The Pulse Oximeter

Name: _____

Date: _____

How to play: For each question, put an X in the box next to the best answer (one answer per question). Get **6 of 8** right to earn the **heart on fire** badge. Take your time - real doctors think before they answer!

1. What does hemoglobin do?☐
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- A. It fights germs in the blood
- B. It carries oxygen inside red blood cells, like a bus with four oxygen seats
- C. It makes the heart beat
- D. It digests food

2. How can the pulse oximeter measure oxygen without taking any blood?☐
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- A. It listens to the blood with a tiny microphone
- B. It smells the skin
- C. It shines red and infrared light through the foot and measures how much gets absorbed
- D. It counts the baby's breaths

3. Why is oxygen-rich blood bright red?☐
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- A. Because of red food coloring in milk
- B. Hemoglobin changes color when it's loaded with oxygen
- C. Because the heart paints it
- D. All blood is exactly the same color

4. A healthy kid's oxygen saturation is usually about...☐
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- A. 50-60%
- B. 70-80%
- C. 95-100%
- D. exactly 200%

5. Why do NICU teams target a LOWER saturation (about 90-95%) for preemies getting extra oxygen?

- ☐ A. To save money on oxygen
- ☐ B. Because too much oxygen can damage a preemie's developing eyes and lungs
- ☐ C. Because the machines can't count higher
- ☐ D. Because preemies prefer it

6. What did huge studies find about aiming for a VERY low target (85-89%)?

- ☐ A. It made babies grow faster
- ☐ B. There were fewer eye problems, but more babies died – so it was too low
- ☐ C. Nothing changed at all
- ☐ D. The babies got taller

7. The desat alarm goes off. What does a great NICU nurse do FIRST?

- ☐ A. Look at the baby
- ☐ B. Turn the alarm volume down
- ☐ C. Call the baby's grandparents
- ☐ D. Restart the computer

8. What happened when 1940s-50s doctors gave preemies as much oxygen as possible?

- ☐ A. All the babies became super strong
- ☐ B. Nothing – oxygen is always safe
- ☐ C. Thousands of babies were blinded by ROP before scientists proved too much oxygen was the cause
- ☐ D. The incubators floated away

Score: _____ / 8

Badge earned? YES / NOT YET (need 6)

LESSON 3 QUIZ

The Monitor

Name: _____

Date: _____

How to play: For each question, put an X in the box next to the best answer (one answer per question). Get **6 of 8** right to earn the **chart** badge. Take your time - real doctors think before they answer!

1. What starts every single heartbeat?☐
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- A. A tiny electrical spark from the heart's own pacemaker cells
- B. A signal from the stomach
- C. The lungs squeezing the heart
- D. Nothing – hearts just move randomly

2. How does the monitor know the heart is beating?☐
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- A. It watches the baby with a camera
- B. Sticky leads on the chest pick up the heart's electrical signals through the skin
- C. A nurse types in each heartbeat
- D. It guesses based on the baby's age

3. Which heart rate is NORMAL for a newborn?☐
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- A. About 30 beats per minute
- B. About 60 beats per minute
- C. About 140 beats per minute
- D. About 500 beats per minute

4. What is apnea of prematurity?☐
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- A. When a preemie's immature brain simply forgets to send the next breathing order
- B. When a baby breathes too fast
- C. A disease of the toes
- D. When a baby sneezes many times in a row

5. In NICU talk, what are “A's and B's”?

- ☐ A. The baby's report card grades
- ☐ B. Apnea (breathing pause) and bradycardia (slow heart rate) spells, which often come together
- ☐ C. Two kinds of baby formula
- ☐ D. The names of the monitor's buttons

6. What is usually the **FIRST** fix when a preemie has an apnea spell?

- ☐ A. Emergency surgery
- ☐ B. A gentle rub on the back or foot to remind the brain to breathe
- ☐ C. A glass of orange juice
- ☐ D. Turning off the monitor

7. What everyday molecule is one of the NICU's best-proven medicines for apnea?

- ☐ A. Sugar
- ☐ B. Salt
- ☐ C. Caffeine — the same molecule as in coffee
- ☐ D. Mint

8. Why do babies breathe and beat so much faster than adults?

- ☐ A. They're always scared
- ☐ B. Small hearts and lungs move less per beat and breath, and fast-growing bodies need lots of oxygen and fuel
- ☐ C. The NICU air is thinner
- ☐ D. They're practicing for races

Score: _____ / 8

Badge earned? YES / NOT YET (need 6)

LESSON 4 QUIZ

Blood Pressure

Name: _____

Date: _____

How to play: For each question, put an X in the box next to the best answer (one answer per question). Get **6 of 8** right to earn the **balloon** badge. Take your time - real doctors think before they answer!

1. What IS blood pressure?

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- A. How much blood a person has
- B. How hard the blood pushes against the walls of the arteries
- C. How fast the heart beats
- D. How red the blood is

2. In “64 over 38,” what is the 64?

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- A. The baby's weight
- B. The diastolic pressure, between beats
- C. The systolic pressure — the peak push during a heartbeat squeeze
- D. The number of heartbeats

3. Why is a newborn's healthy blood pressure so much LOWER than an adult's?

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- A. Newborns are lazy
- B. Smaller body, shorter pipes, smaller pump — smaller pressures do the job
- C. Baby blood is thinner than water
- D. It isn't — babies run 120/80 too

4. What's the famous rule of thumb for a preemie's minimum MAP?

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- A. MAP should be at least 120
- B. MAP should be at least about the baby's age-in-the-womb in weeks
- C. MAP should equal the baby's weight in pounds
- D. MAP doesn't matter for preemies

5. How can doctors get a beat-by-beat blood pressure from a newborn without poking through the skin?

- ☐ A. By slipping a soft tube into a blood vessel of the umbilical cord stump
- ☐ B. By squeezing the baby's foot
- ☐ C. By weighing the baby very precisely
- ☐ D. It's impossible

6. Why is making pee such good news for a sick baby?

- ☐ A. It means the diaper company gets paid
- ☐ B. Kidneys only make urine when they're receiving good blood flow – pee is proof of delivery
- ☐ C. Pee keeps the baby warm
- ☐ D. It isn't good news

7. A preemie's blood pressure is low and the baby looks poorly. Which is NOT one of the team's real tools?

- ☐ A. A careful dose of IV fluid
- ☐ B. A medicine drip like dopamine to strengthen the pump and tighten the vessels
- ☐ C. Hunting for a cause like infection or blood loss
- ☐ D. A tiny cup of hot cocoa

8. Why is steady blood pressure especially important for a very premature baby's BRAIN?

- ☐ A. The brain floats higher when pressure is low
- ☐ B. Its brand-new fragile blood vessels can be starved by low flow or burst and bleed with surges
- ☐ C. Low pressure makes babies forget their names
- ☐ D. It only matters for the feet

Score: _____ / 8

Badge earned? YES / NOT YET (need 6)

LESSON 5 QUIZ

CPAP & the Ventilator

Name: _____

Date: _____

How to play: For each question, put an X in the box next to the best answer (one answer per question). Get **6 of 8** right to earn the **wind** badge. Take your time - real doctors think before they answer!

1. What are alveoli?☐
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- A. The hairs inside your nose
- B. The microscopic air balloons in the lungs where oxygen enters the blood
- C. Tubes that carry food
- D. A type of Italian pasta

2. What does surfactant do?☐
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- A. It makes the lungs smell nice
- B. It coats the wet alveoli like slippery soap so they don't stick shut
- C. It carries oxygen in the blood
- D. It makes babies sleepy

3. Why do very premature babies get RDS (respiratory distress syndrome)?☐
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- A. They breathe too much dust
- B. Their lungs don't make enough surfactant yet — that comes near the end of pregnancy
- C. Their noses are too small
- D. They catch it from other babies

4. Blowing up a balloon teaches you the secret of CPAP. What is it?☐
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- A. Balloons are stronger than lungs
- B. The first puff from empty is the hardest — CPAP keeps the lungs partly inflated so no breath ever starts from empty
- C. You should always breathe into a balloon
- D. CPAP fills the baby with helium

5. What does a ventilator do that CPAP doesn't?

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A. It actually delivers the breaths, through a tube in the windpipe, with size, speed, and oxygen all precisely dialed in

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B. It rocks the baby to sleep

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C. It feeds the baby milk

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D. It keeps the baby warm

6. What's the NICU's secret weapon given BEFORE a premature birth?

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A. A lucky charm in the delivery room

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B. A steroid medicine given to the mother that tells the baby's lungs to mature faster

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C. Extra blankets

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D. Loud music to wake the lungs up

7. What happened to Patrick Bouvier Kennedy, the President's son, in 1963?

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A. He was cured with surfactant

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B. He died of RDS at 39 hours old, because no treatment existed yet – even for a president's child

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C. He grew up to be a doctor

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D. He was the first baby on CPAP

8. Who discovered that RDS lungs were missing surfactant?

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A. Dr. Mary Ellen Avery, in 1959

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B. An astronaut

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C. The inventor of the balloon

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D. Nobody knows how RDS works

Score: _____ / 8

Badge earned? YES / NOT YET (need 6)

LESSON 6 QUIZ

Feeding Tubes & IV Lines

Name: _____

Date: _____

How to play: For each question, put an X in the box next to the best answer (one answer per question). Get **6 of 8** right to earn the milk badge. Take your time - real doctors think before they answer!

1. Why CAN'T a baby born at 26 weeks drink from a bottle?☐

A. It doesn't like milk

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B. The suck-swallow-breathe rhythm isn't wired into the brain until about 32-34 weeks – milk would go into the lungs

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C. Its mouth is too big

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D. Bottles aren't allowed in hospitals

2. What is an NG tube?☐

A. A tube that plays music

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B. A soft tube from the nose down to the stomach for delivering milk

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C. A wire that measures brain waves

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D. A straw for the nurse's coffee

3. What is TPN?☐

A. A TV channel for babies

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B. Complete nutrition – sugar, protein, fat, vitamins, minerals – given directly into the bloodstream

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C. A special warm blanket

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D. A brand of diapers

4. Trophic feeds – the first tiny 1-2 mL feeds – are mainly for...☐

A. filling the baby up

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B. gently waking up and exercising the gut, like practice laps

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C. keeping the syringe company

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D. making the baby taller

5. What is NEC?☐

A. A dangerous disease where parts of a preemie's immature intestine become inflamed and start to die

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B. A machine that warms milk

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C. A type of vitamin

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D. The name of the night nurse

6. Why is breast milk described as MEDICINE for preemies?☐

A. It's warmer than formula

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B. It carries antibodies, is gentle on immature guts, and substantially lowers the risk of NEC

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C. It's cheaper

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D. It makes babies sleep through the night

7. A 'micro-preemie' weighing 500 grams weighs about as much as...☐

A. a bicycle

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B. two sticks of butter

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C. a watermelon

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D. a golden retriever

8. Why do nurses give a preemie a pacifier during tube feeds?☐

A. To keep the baby quiet during TV time

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B. It's sucking practice – safe rehearsal for real feeding later

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C. To measure the mouth

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D. It holds the feeding tube in place

Score: _____ / 8

Badge earned? YES / NOT YET (need 6)

LESSON 7 QUIZ

The Bili Lights

Name: _____

Date: _____

How to play: For each question, put an X in the box next to the best answer (one answer per question). Get **6 of 8** right to earn the **blue heart** badge. Take your time - real doctors think before they answer!

1. Where does bilirubin come from?☐
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- A. From eating too many bananas
- B. It's the yellow leftover from recycling worn-out red blood cells
- C. The hospital lights make it
- D. It comes from baby shampoo

2. Why do so many newborns get jaundiced?☐
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- A. They drink too much milk
- B. Extra red cells from womb-life all get recycled at once, while the brand-new liver is still slow at flushing bilirubin out
- C. They're cold
- D. Jaundice is caused by crying

3. How does blue light treat jaundice?☐
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- A. It heats the skin until bilirubin melts
- B. It scares the bilirubin away
- C. Bilirubin absorbs blue light and twists into new shapes the body can flush out in pee and poop – no liver needed
- D. It turns the skin blue so the yellow doesn't show

4. Why does the baby wear eye shields under the bili lights?☐
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- A. To look cool (only partly!) – mainly because the light is intensely bright and the eyes deserve protection
- B. So the baby can't see the doctors
- C. To keep the eyes warm
- D. The shields measure brain waves

5. What is kernicterus?

- ☐ A. A funny name for the hiccups
- ☐ B. Permanent brain damage from extremely high bilirubin – preventable by checking and treating jaundice in time
- ☐ C. A type of German pastry
- ☐ D. The machine that measures yellowness

6. How was phototherapy discovered?

- ☐ A. A computer invented it
- ☐ B. Nurse Jean Ward noticed her sunshine-loving babies were less yellow – except where the blanket covered them
- ☐ C. It was found in an ancient book
- ☐ D. A baby invented it by rolling into the sun

7. Which is a real jaundice treatment besides the lights?

- ☐ A. Feeding lots of milk so the baby poops the bilirubin out
- ☐ B. Painting the baby blue
- ☐ C. Keeping the baby awake
- ☐ D. Lemon juice

8. Why do doctors judge a bilirubin level using the baby's age IN HOURS?

- ☐ A. Doctors love complicated charts
- ☐ B. Because a level that's normal at 4 days old can be dangerous at 12 hours old – how fast it's rising matters
- ☐ C. Because babies can't tell time
- ☐ D. The machines only count hours

Score: _____ / 8

Badge earned? YES / NOT YET (need 6)

LESSON 8 QUIZ

Seeing Inside

Name: _____

Date: _____

How to play: For each question, put an X in the box next to the best answer (one answer per question). Get **6 of 8** right to earn the **magnifying glass** badge. Take your time - real doctors think before they answer!

1. Why do bones show up WHITE on an X-ray?☐
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- A. Bones glow in the dark
- B. Dense bone blocks the X-rays, leaving a bright shadow on the picture
- C. The machine paints them white
- D. Bones are cold

2. How does an ultrasound make a picture?☐
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- A. It uses a tiny camera on a wire
- B. It sends super-high sounds into the body and times the echoes bouncing off the organs – like a dolphin
- C. It uses X-ray light
- D. It reads the baby's mind

3. Why is the fontanelle (soft spot) so useful to NICU doctors?☐
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- A. It's where the baby's thoughts come out
- B. Sound can't pass through skull bone – but the bone-free soft spot is a perfect window for brain ultrasounds
- C. It's the best spot for stickers
- D. It keeps the hat on

4. Why does every baby born before about 30-32 weeks get a head ultrasound in the first week – even if nothing seems wrong?☐
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- A. To take a cute first photo
- B. Because brain bleeds (IVH) in preemies usually happen early and often silently – screening is the only way to know
- C. To measure hat size
- D. It's just tradition

5. Before birth, why does blood take a detour AROUND the lungs?

- ☐ A. The lungs are on vacation
- ☐ B. The lungs are folded up and fluid-filled – the placenta does the oxygen job, so the ductus arteriosus lets blood skip them
- ☐ C. Blood is afraid of the lungs
- ☐ D. It doesn't – unborn babies breathe air

6. What is a PDA?

- ☐ A. A ductus arteriosus that stays stuck open after birth, letting blood flood the lungs – common in preemies
- ☐ B. A tiny robot in the heart
- ☐ C. A kind of baby monitor
- ☐ D. A vitamin deficiency

7. Which everyday medicine can help close a PDA?

- ☐ A. Cough syrup
- ☐ B. Ibuprofen – the same medicine grown-ups take for headaches
- ☐ C. Sunscreen
- ☐ D. Toothpaste

8. Why do doctors reach for ultrasound instead of X-ray whenever it can answer the question?

- ☐ A. Ultrasound machines are prettier
- ☐ B. Ultrasound uses no radiation and gives live pictures, so it can be repeated at the bedside as often as needed
- ☐ C. X-rays are too loud
- ☐ D. There's no real reason

Score: _____ / 8

Badge earned? YES / NOT YET (need 6)

FINAL CHALLENGE

Your NICU Rounds

Name: _____

Date: _____

How to play: For each question, put an X in the box next to the best answer (one answer per question). Get **9 of 12** right to earn the **stethoscope** badge. Take your time - real doctors think before they answer!

1. Bed 1: A baby born at 26 weeks keeps drifting cold even in a normal crib. Why can't this baby hold its own temperature?

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A. The baby doesn't try hard enough

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B. Thin skin, no insulating fat, no shivering, and almost no brown fat – plus lots of skin surface for such a tiny body

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C. The hospital is too big

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D. Babies are cold-blooded like lizards

2. Bed 2: The pulse oximeter on a 27-weeker getting extra oxygen reads a steady 100%. Your move?

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A. Perfect – leave everything as is

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B. Turn the oxygen UP to be safe

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C. Turn the extra oxygen DOWN a bit – sitting at 100% on oxygen risks the baby's eyes and lungs

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D. Remove the probe, it must be broken

3. Bed 3: The monitor alarms – flat breathing line for 20 seconds, heart rate dipping to 80. The nurse rubs the baby's back and everything recovers. What just happened?

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A. A nightmare

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B. A classic A-and-B spell: apnea of prematurity with bradycardia – the immature brain forgot to breathe

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C. The monitor was unplugged

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D. The baby was holding its breath on purpose

4. Bed 4: A 25-weeker's MAP is reading 22 (below the born-at-25-weeks rule of thumb). Before treating, what do you check FIRST?

- ☐ A. The baby itself: pee output, warmth, cap refill, blood tests — is blood actually being delivered?
- ☐ B. Whether the number is divisible by two
- ☐ C. The weather
- ☐ D. Nothing — always start dopamine immediately

5. Bed 5: A baby born at 29 weeks is grunting, flaring its nostrils, with skin sucking in between the ribs. Chest X-ray shows hazy lungs. Diagnosis and treatment?

- ☐ A. A cold — wait it out
- ☐ B. RDS from missing surfactant — support with CPAP, and give surfactant down a tube if needed
- ☐ C. Hiccups — apply juice
- ☐ D. Too much exercise

6. Bed 6: Why is the 26-weeker in bed 6 fed 2 mL of milk through a nose tube instead of a bottle?

- ☐ A. Bottles are sold out
- ☐ B. The suck-swallow-breathe rhythm isn't wired until about 32-34 weeks — bottle milk could end up in the lungs
- ☐ C. The baby prefers tubes
- ☐ D. Nurses find bottles boring

7. Bed 6, continued: The nurse notices this baby's belly is newly swollen and the last feed never emptied from the stomach. Why does everyone move fast?

- ☐ A. Swollen bellies are good luck
- ☐ B. These can be early signs of NEC, the dangerous gut disease of preemies — stop feeds, image the belly, act quickly
- ☐ C. The baby is just chubby
- ☐ D. It means the baby is ready for pizza

8. Bed 7: A 3-day-old term baby is noticeably yellow. The skin meter reads high, blood test confirms. What's the treatment?

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- A. Blue-light phototherapy (plus plenty of milk) – light twists bilirubin into flushable shapes
- B. A warm bath to wash the yellow off
- C. Waiting a month
- D. Yellow paint to match

9. Bed 8: A baby was born last night at 27 weeks. Nothing seems wrong neurologically. Does this baby still need a head ultrasound this week?

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- A. No – no symptoms, no scan
- B. Yes – IVH brain bleeds in very early preemies usually happen in the first days and are often silent; screening through the fontanelle is routine
- C. Only if the parents insist
- D. Head ultrasounds are for adults

10. Bed 9: The echo on a tiny preemie shows the ductus arteriosus still wide open (a PDA), flooding the lungs. Which is NOT one of your real options?

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- A. Watch and wait – many close on their own
- B. A medicine like ibuprofen to nudge the vessel shut
- C. A tiny plug threaded in through a leg vein
- D. Asking the baby to concentrate very hard on closing it

11. Rounds pop-quiz from your attending: Why does the NICU target 90-95% oxygen saturation for preemies instead of 100%?

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- A. The machines get tired above 95
- B. Because huge studies showed too much oxygen damages eyes and lungs, while too little costs lives – 90-95% balances both dangers
- C. Insurance rules
- D. Round numbers are unlucky

12. Last stop: A mother is doing kangaroo care, her 30-weeker asleep on her chest, monitor numbers rock-steady. A visitor asks why 'just cuddling' is allowed in intensive care. Your best answer?

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A. It's not allowed – sound the alarm

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B. Skin-to-skin contact is real treatment: it holds the baby's temperature, steadies heart rate and breathing, and supports feeding and bonding

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C. The nurses needed a break

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D. It's for photos only

Score: _____ / 12

Badge earned? YES / NOT YET (need 9)